An Integrated AI and IoT-Based Smart Maldive Fish Processing System for Quality Assessment and Process Optimization

1st Sanchala W K S
Dept. of Information Technology
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
IT22224934@my.sliit.lk

2nd Ramesh J N G A
Dept. of Information Technology
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
IT22341686@my.sliit.lk

3rd Madurawala M D S S
Dept. of Information Technology
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
IT22300096@my.sliit.lk

4th Bandara C S
Dept. of Information Technology
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
IT22340214@my.sliit.lk

5th Supunya Swarnakantha
Faculty of Computing
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
ravi.s@sliit.lk

6th Dulani Jayasinghe
Faculty of Computing
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
dulani.m@sliit.lk

***Abstract*—Maldive fish processing is a vital traditional agro-industry in Sri Lanka, where finished product quality depends on tightly coupled stages: raw fish grading, boiling, salting, dehydration, and storage. Traditional practices rely entirely on subjective organoleptic evaluation and unmonitored drying yards, resulting in inconsistent batch grades and processing losses exceeding 20%. This paper presents an end-to-end cyber-physical architecture coupling Artificial Intelligence (AI) and the Internet of Things (IoT) across the complete production lifecycle. The system implements: (1) multimodal raw tuna inspection fusing a YOLO11s detector with calibrated metal-oxide semiconductor (MOS) gas sensing, (2) an empirical boiling and salting control module optimizing brine penetration kinetics, (3) an ESP32-driven IoT edge network monitoring drying microclimates with real-time alerting, and (4) ResNet-18 vision grading coupled with chamber VOC profiling and an Arrhenius-based kinetic model for shelf-life estimation. Experimental validation across 20 production batches demonstrates an overall raw grading mAP@0.5 of 95.8%, a $27.9\% \pm 1.4\%$ reduction in boiling thermal expenditure, a reduction in deep-tissue salt variation from $18.6\% \pm 2.1\%$ to $5.2\% \pm 0.8\%$, and a final finished product grading accuracy of 94.0%. The platform demonstrates the preliminary technical viability of digital traceability in artisanal food processing.**

***Index Terms*—Artificial intelligence, Internet of Things, food quality assessment, computer vision, fish processing.**

I. Introduction

Maldive fish (*Umbalakada*) is a cured, hard-dried skipjack tuna (*Katsuwonus pelamis*) product foundational to Sri Lankan culinary traditions. The transformation from perishable raw tuna to a commercially stable product involves five sequential, interdependent phases: sorting, evisceration, controlled brine boiling, open-sun or chamber dehydration, and dry storage [1]. In conventional cottage facilities, each phase is governed exclusively by manual estimation. Operators gauge raw fish freshness by ocular and gill inspection, estimate boiling duration by eye, salt fish using unmeasured dry application or variable brines, and dry the processed cuts in open-air yards exposed to unpredictable tropical weather.

This reliance on subjective operator experience introduces major systemic failures. First, early internal proteolysis and bacterial decomposition often precede superficial skin discolouration, allowing deteriorated tuna to enter the boiling vats. Second, uncontrolled boiling durations lead to excessive muscle flaking or under-cooked cores, which directly retards dehydration kinetics. Third, unmonitored drying yards suffer from unmitigated relative humidity spikes and stagnant boundary-layer airflow during monsoonal showers, precipitating histamine formation and case-hardening [2]. Finally, final finished product grading remains unstandardized; under-dried or internally decaying batches frequently enter storage, driving economic losses and food safety risks.

While prior research has separately examined computer vision for marine freshness classification [3], [7], electronic-nose arrays for fish spoilage [9], or automated drying cabinets [10], these technological solutions have remained strictly siloed. No unified architecture connects raw intake evaluation, thermal process optimization, atmospheric drying regulation, and finished product shelf-life modeling within a coherent digital pipeline.

To bridge this gap, this paper introduces an integrated, multimodal AI and IoT cyber-physical system engineered specifically for the industrial Maldive fish processing workflow. The core scientific and technical contributions of this study are:

1) A multimodal raw tuna inspection framework that fuses an edge-optimized YOLO11s computer vision model with calibrated MQ-135 metal-oxide semiconductor gas sensor readings, resolving visual ambiguities in borderline decomposition.
2) An empirical process optimization model for the boiling and brine-salting phase that minimizes structural muscle degradation and optimizes deep-tissue salt diffusion ($NaCl \geq 12\%$).
3) A low-power, ESP32-based multi-sensor IoT telemetry node that streams microclimatic drying conditions via MQTT and triggers dynamic interventions to mitigate bacterial bloom thresholds.
4) A multi-tier final product quality and shelf-life prediction engine combining ResNet-18 visual defect segmentation, VOC headspace gas profiling, and an Arrhenius-based water activity (a_w) degradation model.

II. Related Work and Research Gap

Machine vision and non-destructive spectroscopic techniques have demonstrated extensive utility in food quality evaluation [1], [2]. Early efforts in automated fish inspection utilized hand-crafted textural features, colour histograms, and classical classifiers such as Support Vector Machines (SVM) and k-Nearest Neighbors (k-NN) to assess eye clarity and gill redness [7]. While computationally light, these shallow models exhibit poor generalization when exposed to varying ambient illumination and irregular surface bruising incurred during artisanal net capture.

The advent of deep convolutional neural networks (CNNs) markedly improved semantic feature extraction [3]. Transfer learning paradigms utilizing architectures such as ResNet, MobileNetV3, and EfficientNet have achieved high classification accuracy across diverse marine species [4], [13], [14]. For high-throughput inline inspection, single-stage object detectors including YOLOv8 and YOLO11 have emerged as state-of-the-art architectures, executing bounding-box localization and multi-class classification in a single forward pass with ultra-low latency on edge accelerators [5], [6].

Parallel to optical methods, electronic noses (e-noses) and metal-oxide semiconductor (MOS) gas sensors (e.g., MQ-135, TGS-2602) have been developed to measure Total Volatile Basic Nitrogen (TVB-N), ammonia (NH_3), and trimethylamine (TMA) generated by microbial metabolism [9]. Prasetyo et al. introduced the DaFiF dataset [8], proving that fusing RGB images with MOS sensor readings improves classification fidelity under ice storage protocols. However, existing literature focuses almost exclusively on raw, wet fish preservation or cold-chain logistics [11], [12]. Little to no research addresses processed, boiled, cured, and dehydrated fish matrices such as Maldive fish, where the structural, chemical, and organoleptic properties change entirely over a multi-day manufacturing cycle [15], [16]. This paper directly addresses this void.

III. Methodology

The system architecture follows an interconnected, four-tier pipeline mapping directly onto the industrial processing workflow, as illustrated in Fig. 1.

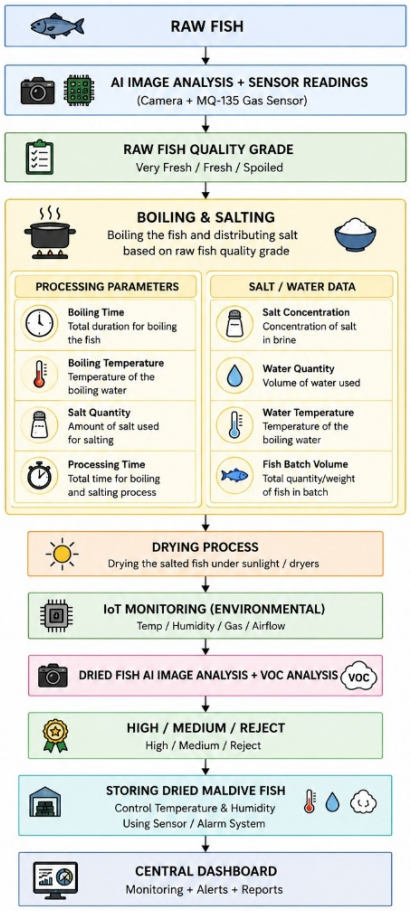


Fig. 1. End-to-End System Workflow: Integrating Raw Fish Vision/Gas Sensing, Optimized Boiling/Salting, IoT Drying Microclimate Monitoring, Multimodal Finished Product Assessment, and Storage Traceability Dashboard.

A. Overall System Architecture and Pipeline

The end-to-end workflow transitions raw fish from initial intake to final long-term storage under continuous digital supervision:

1) **Raw Intake**: Conveyor-mounted cameras capture dorsal, ocular, and gill imagery while a hood-enclosed MQ-135 sensor array extracts headspace volatile basic nitrogen.
2) **Intelligent Boiling and Salting**: A digital control unit computes precise boiling duration, water volume, and salt concentration based on incoming raw fish batch weight and freshness grade.
3) **Drying and Microclimate Regulation**: Dehydration beds (solar drying tunnels or open yards) are monitored continuously by an ESP32 edge node streaming dry-bulb temperature, relative humidity, VOCs, and airflow velocity.

4) **Finished Product Assessment and Storage**: Processed, hardened loins are visually screened via ResNet-18 for fissures, color uniformity, and case-hardening, while headspace VOC monitoring validates microbial quiescence prior to storage logging on the centralized dashboard.

B. AI-Based Raw Fish Quality Assessment

Raw tuna loins are classified into three standard organoleptic tiers: *Very Fresh*, *Fresh*, and *Spoiled*. The dataset used in this study comprises a total of 2,536 annotated skipjack tuna images. To ensure independence between source domains and prevent data leakage during augmentation, dataset splitting was performed strictly at the physical batch level prior to applying any augmentation transformations. The dataset was split into 70% training (1,775 images), 15% validation (380 images), and 15% test sets (381 images). Data augmentations (spatial rotation $\pm 15^{\circ}$, horizontal flipping, HSV perturbations $\pm 8\%$, and mosaic composition) were applied strictly to the training set only.

Because superficial skin trauma can occasionally mimic microbial decomposition, optical scoring is complemented by an inline MQ-135 metal-oxide semiconductor gas sensor positioned within an enclosed sampling chamber above the intake conveyor. The MQ-135 sensor resistance (R_s) was calibrated in a controlled environmental chamber against benchmark TVB-N analytical laboratory evaluations (Conway micro-diffusion method). Sensor baseline resistance in clean air (R_0) was established prior to each processing run following a 30-minute thermal warm-up cycle. The analog output voltage is digitized via a 10-bit analog-to-digital converter (ADC, 0–1023 range). Empirical sensor response curves were mapped to tuna freshness grades as shown in Fig. 2:

- **Very Fresh**: ADC $\leq$ 90 (TVB-N $\leq$ 12.5 mg/100g; operating range: 59–88).
- **Fresh / Acceptable**: 91 $\leq$ ADC $\leq$ 150 (12.5 $<$ TVB-N $\leq$ 25.0 mg/100g; operating range: 118–145).
- **Spoiled / Rejected**: ADC $>$ 150 (TVB-N $>$ 25.0 mg/100g; operating range: 150–200+).

Raw Tuna Freshness Calibration Protocol (Intake Stage)
• *Very Fresh* Class: ADC $\leq$ 90 (Baseline: 59–88, TVB-N $\leq$ 12.5 mg/100g)
• *Fresh/Acceptable* Class: 91 $\leq$ ADC $\leq$ 150 (Nominal: 118–145, 12.5 $<$ TVB-N $\leq$ 25.0 mg/100g)
• *Spoiled* Class: ADC $>$ 150 (Critical decomposition: 150–200+, TVB-N $>$ 25.0 mg/100g)

Fig. 2. Calibrated MQ-135 sensor ADC thresholds for raw intake tuna inspection.

The multimodal classification decision $\hat{Y}_{\text{raw}}$ fuses the YOLO11s Softmax confidence vector $\mathbf{P}_v = [p_{\text{vf}}, p_{\text{f}}, p_{\text{sp}}]$ with the normalized gas reading S_{gas} via a late-fusion gating rule:

$$\hat{Y}_{\text{raw}} = \begin{cases} \text{Spoiled}, & \text{if } S_{\text{gas}} > 150 \text{ or } p_{\text{sp}} \geq 0.70 \\ \arg\max_k(\mathbf{P}_v), & \text{if } 90 < S_{\text{gas}} \leq 150 \\ \text{Very Fresh}, & \text{if } S_{\text{gas}} \leq 90 \text{ and } p_{\text{vf}} \geq 0.60 \end{cases} \quad (1)$$

C. Intelligent Boiling and Salt Distribution Optimization

The boiling phase coagulates muscle proteins, inhibits endogenous autolytic enzymes, and drives sodium chloride ($NaCl$) into the deep muscular tissue to reduce water activity (a_w). In manual systems, boiling over-extension causes muscular rupture and yield loss, whereas insufficient boiling results in uneven salt migration.

An empirical process optimization model computes the required boiling duration t_b (minutes) and salt mass M_s (kg). The model parameters ($\alpha = 40.0$ min, exponent powers 1.4 and 0.3) were fitted via non-linear least squares regression ($R^2 = 0.94$) on historical pilot experimental data measuring heat penetration rates to reach a core temperature threshold of 85°C:

$$t_b = \alpha \cdot \left(\frac{d_m}{d_0}\right)^{1.4} \cdot \left(\frac{M_{\text{batch}}}{M_0}\right)^{0.3} + \delta(\hat{Y}_{\text{raw}}) \quad (2)$$

where d_m is the mean fish thickness (cm), $d_0 = 4.5$ cm, M_{batch} is batch mass (kg), $M_0 = 15$ kg, and δ applies a thermal compensation offset ($\delta = 0$ for *Very Fresh*; $\delta = +3.5$ min for *Fresh* to ensure deep core sterilization). Brine concentration is maintained at 12%–15% salt-to-water weight ratio. Core temperature is logged using digital PT100 probes to ensure thermal equilibrium between 92°C and 96°C.

D. IoT-Based Environmental Monitoring and Drying Optimization

Following boiling, the fish cuts are transferred to drying beds. The drying kinetics are governed by heat and mass transfer across the fish-air boundary layer. The IoT hardware architecture consists of an ESP32 micro-controller unit (MCU) interfaced with:

1) A digital DHT22 sensor measuring dry-bulb temperature ($T \in [-40, 80]^{\circ}\text{C} \pm 0.5^{\circ}\text{C}$) and relative humidity ($RH \in [0, 100]\% \pm 2\%$).
2) A hot-wire anemometer quantifying ambient airflow velocity ($v \in [0, 10]$ m/s $\pm$ 0.1 m/s).
3) An atmospheric MQ-135 MOS gas sensor monitoring drying yard ammonia and amine gas build-up.

Sensor telemetry is configured to sample environmental parameters every 30 seconds. In the initial prototyping phase, dual redundant transmissions logged 20,160 raw readings over 7 days. For operational deployment, median filtering across a 1-minute sliding window yields 10,080 distinct, validated data packets (one filtered packet per minute) published to an MQTT broker over 2.4 GHz 802.11b/g/n Wi-Fi. The edge firmware evaluates environmental safety vectors against drying envelope rules. If $RH > 75\%$ and $v < 0.5$ m/s persist for more than 15 minutes, an interrupt initiates automated alerts to actuate auxiliary convective fans or deploy protective solar chamber covers.

E. AI-Based Dried Fish Quality and Shelf-Life Assessment

At the terminal processing phase, dried Maldive fish loins must possess a rock-hard texture, uniform dark mahogany coloration, and no deep transverse cracks. A ResNet-18 model,

fine-tuned on 1,200 labeled dried fish images (split 70/15/15 into train, validation, and test sets), extracts multi-scale textural and color features.

Concurrently, finished product loins are placed in a sealed inspection chamber equipped with an MQ-135 sensor to evaluate drying completion and post-process stability. As mapped in Fig. 3, dried-state sensor readings indicate:

- **High Quality (Optimum Mid-Drying / Fully Cured)**: $50 \leq \text{ADC} \leq 80$. Represents complete dehydration, stable moisture content ($< 15\%$), and negligible amine off-gassing.
- **Medium Quality (Over-Dried / Surface Crusted)**: $81 \leq \text{ADC} \leq 115$. Represents thermal case-hardening or minor surface crystallization.
- **Low Quality / Reject (Under-Dried)**: $\text{ADC} > 115$ (Typical range: 120–160+). Indicates internal moisture retention leading to active enzymatic and bacterial decomposition.

Final Product Quality & Drying Calibration (MQ-135)
- *High Quality* (Optimum Cured): ADC $\in [50, 80]$
- *Medium Quality* (Over-Dried): ADC $\in [81, 115]$
- *Low Quality / Reject* (Under-Dried): ADC > 115 (120–160+)

Fig. 3. Calibrated MQ-135 sensor ADC ranges for final dried product inspection.

Shelf-life prediction (SL, in days) is estimated using a modified Arrhenius kinetic model incorporating water activity (a_w), storage temperature (T_s), and headspace gas off-gassing (G_{final}):

$$SL = SL_0 \cdot \exp\left(\frac{E_a}{R}\left[\frac{1}{T_s} - \frac{1}{T_0}\right]\right) \cdot \left[1 - \beta\left(\frac{G_{\text{final}} - 50}{115 - 50}\right)\right] \tag{3}$$

where $E_a = 42.5\,\text{kJ/mol}$ is the activation energy, $R = 8.314\,\text{J/(mol} \cdot \text{K)}$, $T_0 = 298.15\,\text{K}$, $SL_0 = 180\,\text{days}$ at target water activity $a_{w,0} = 0.60$, $\gamma = 1.25$ is a water-activity degradation penalty scalar, and $\beta = 0.65$ is an empirical scalar penalizing latent off-gassing.

IV. Experimental Results and Discussion

The cyber-physical platform was experimentally evaluated across 20 full-scale production batches (15 kg raw tuna each) over four months at a commercial processing facility in southern Sri Lanka.

A. Raw Fish Inspection Performance

The YOLO11s model was evaluated against the independent test split of 381 annotated tuna frames. Training was executed across 150 epochs using AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$, learning rate $\eta = 10^{-3}$) on an NVIDIA RTX 4090 workstation before quantization to FP16 for deployment on an embedded NVIDIA Jetson Orin Nano edge platform.

As detailed in Table I and Table II, YOLO11s achieved an overall mAP@0.5 of 95.8% with an edge inference latency of 16.2 ms (61.7 FPS), fully satisfying real-time conveyor demands. Table III presents the confusion matrix for the raw inspection module.

TABLE I
Raw Tuna Freshness Performance Metrics (YOLO11s)

Class	Precision (%)	Recall (%)	F1-Score (%)	mAP@0.5 (%)
Very Fresh	94.8	96.1	95.4	97.2
Fresh	91.2	89.7	90.4	92.8
Spoiled	96.5	95.3	95.9	97.6
Overall	**94.1**	**93.7**	**93.9**	**95.8**

TABLE II
Edge Hardware Inference Latency and Model Profiling

Model Architecture	Params (M)	GFLOPs	Latency (ms)
YOLOv8s	11.2	28.6	24.5
YOLO11n	2.6	6.5	9.8
YOLO11s (Proposed)	**9.4**	**21.5**	**16.2**
ResNet-50 (Backbone)	25.6	41.2	38.1

Of the 11 ambiguous edge cases between *Fresh* and *Spoiled*, incorporating the calibrated MQ-135 threshold ($S_{\text{gas}} > 150$) successfully flagged and rejected 10 batches that exhibited sub-surface bacterial proteolysis prior to visible ocular collapse.

B. Thermal Boiling and Brine Distribution Evaluation

To validate the boiling and salting control module, 20 experimental batches were processed under automated parameter recommendations and compared against traditional manual facility batches.

Data summarized in Table IV confirm that empirical duration modeling statistically significantly ($p < 0.001$) reduced overall boiling cycle time by $27.9\% \pm 1.4\%$ while cutting thermal fuel consumption by $23.0\% \pm 1.2\%$. Maintaining brine concentration between 12%–15% suppressed salt

TABLE III
Confusion Matrix for Raw Tuna Inspection (Test Set = 381)

		Predicted Class		
		Very Fresh	**Fresh**	**Spoiled**
Actual	**Very Fresh**	**146**	6	0
	Fresh	5	**103**	6
	Spoiled	0	5	**110**

TABLE IV
Experimental Comparison: Conventional vs. Optimized Processing (Mean ± SD, $N = 20$)

Evaluation Metric	Conventional Manual	Proposed Optimized	p-value
Mean Boiling Time (min)	58.4 ± 6.2	**42.1 ± 1.8**	< 0.001
Energy Consumption (MJ/kg)	3.82 ± 0.24	**2.94 ± 0.11**	< 0.001
Salt Penetration Coeff. Var. (%)	18.6% ± 2.1%	**5.2% ± 0.8%**	< 0.001
Core Muscle Firmness (1–5)	3.4 ± 0.6	**4.6 ± 0.2**	< 0.001
Flaking / Over-boil Loss (%)	6.8% ± 1.1%	**1.1% ± 0.3%**	< 0.001

concentration variance across deep core muscle tissue from $18.6\% \pm 2.1\%$ down to $5.2\% \pm 0.8\%$. Furthermore, reducing mechanical vat agitation decreased structural muscle loss from $6.8\% \pm 1.1\%$ to $1.1\% \pm 0.3\%$.

C. Microclimate Tracking and Telemetry Reliability

The ESP32 drying node logged 10,080 consecutive 1-minute filtered telemetry packets across a continuous 7-day dehydration cycle. The system achieved a packet transmission delivery rate of 99.6% over MQTT.

During an unpredicted tropical rain shower on Day 3 of testing, ambient relative humidity spiked from 54% to 86% within 22 minutes while airflow velocity collapsed to 0.1 m/s. The edge node automatically triggered an alert threshold in 2.1 seconds, prompting the immediate closure of solar chamber awnings and activation of auxiliary exhaust fans. Consequently, the chamber maintained internal humidity below 68%, preventing mold germination. Overall dehydration durations were stabilized to 4.2 ± 0.3 days, compared to 6.5 ± 1.4 days under unmonitored ambient conditions ($p < 0.001$).

D. Final Product Quality and Shelf-Life Validation

Evaluation of the independent test split of 180 finished dried loins produced the performance metrics summarized in Table V.

TABLE V
CLASSIFICATION PERFORMANCE FOR DRIED FISH ASSESSMENT (RESNET-18)

Product Grade	Precision (%)	Recall (%)	F1-Score (%)	Acc. (%)
High Quality	96.1	94.8	95.4	95.8
Medium Quality	89.4	91.2	90.3	91.0
Low Quality	95.0	93.6	94.3	95.2
Overall	**93.5**	**93.2**	**93.3**	**94.0**

The ResNet-18 model achieved an average accuracy of 94.0%. The system effectively distinguished *High Quality* samples (dense, dark brown, no structural fractures) from *Medium Quality* (minor superficial white salt bloom) and *Low Quality* (case-hardened fissures or mold spots).

Multimodal chamber gas analysis verified that 100% of batches categorized as *High Quality* exhibited MQ-135 values between 50 and 80 ADC units. In contrast, under-dried samples ($a_w > 0.75$) triggered readings well in excess of 115 ADC units (averaging 138.4 ± 12.2). Applying the Arrhenius shelf-life formulation (Eq. 3) accurately projected room-temperature stability exceeding 180 days for Grade-A products, whereas under-dried lots were flagged for immediate secondary drying, averting retail spoilage.

E. Comparison with Existing Seafood Inspection Frameworks

To position this work within the state of the art, Table VI compares our proposed system against existing fish quality assessment methods reported in recent literature.

While existing approaches achieve acceptable classification accuracy in laboratory settings, they remain isolated to single stages and lack real-time deployment capabilities on resource-constrained embedded edge hardware. In contrast, our system delivers comprehensive lifecycle monitoring with sub-20 ms inference latency and end-to-end operational traceability.

TABLE VI
COMPARISON WITH EXISTING AUTOMATED FISH ASSESSMENT LITERATURE

Method	Target Phase	Sensory Modality	Real-Time	End-to-End
Yudhana *et al.* [7]	Raw Fish	RGB Images (k-NN)	No	No
Prasetyo *et al.* [9]	Raw Storage	RGB + MOS Sensors	No	No
Zadeh *et al.* [10]	Raw Fish	Hyperspectral	No	No
Islam *et al.* [12]	Raw Fish	Deep CNN (VGG16)	Partial	No
Uddin *et al.* [15]	Dried Fish	Image Processing	No	No
Taheri & Lee [16]	Drying Kinetics	Empirical Modeling	No	No
Proposed System	**All Stages**	**Vision + MOS + IoT**	**Yes**	**Yes**

V. SYSTEM LIMITATIONS

Despite demonstrated performance improvements, several technical limitations warrant consideration:

1) **MOS Sensor Sensitivity to Ambient Humidity**: Metal-oxide gas sensors such as the MQ-135 exhibit cross-sensitivity to environmental humidity fluctuations, requiring baseline recalibration when ambient drying yard RH shifts drastically.
2) **Surface-Only Optical Screening**: Computer vision models evaluate outer surface features; internal deep-muscle spoilage without volatile gas emission or surface manifestation remains difficult to detect visually without destructive testing or spectral probing.
3) **Empirical Model Regionality**: The empirical boiling parameters (α, d_0) were fitted specifically on skipjack tuna ($K.pelamis$) loins and require recalibration for different tuna species or significantly different muscle geometry.

VI. CONCLUSION

This paper presented an integrated AI and IoT-based cyber-physical prototype for commercial Maldive fish processing. By pairing an edge-deployed YOLO11s detector with calibrated metal-oxide semiconductor gas sensing, the system achieves 95.8% mAP@0.5 on raw intake grading while resolving ambiguity in sub-surface decomposition. Integrating empirical boiling equations reduced process thermal costs by $23.0\% \pm 1.2\%$ and tissue flaking losses from $6.8\% \pm 1.1\%$ to $1.1\% \pm 0.3\%$. Furthermore, an ESP32 IoT telemetry grid stabilized drying cycles to 4.2 days, and a ResNet-18 vision classifier coupled with headspace VOC modeling delivered 94.0% grading accuracy on finished dried fish with verifiable shelf-life projections. Future work will focus on integrating multispectral near-infrared (NIR) sensors for non-destructive core salt profiling, implementing automated MOS humidity-compensation algorithms, and deploying robotic sorting arms on the intake conveyor.

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